is low. Further, the Commission will consider the state of the market and the general readiness of the competing consolidator infrastructure in determining whether to approve a national market system plan amendment that will effectuate a cessation of the operation of the existing exclusive SIPs.

(i) Factors

a. Barriers to Entry

The first factor that will affect the number of competing consolidators is the barriers to entry. Potential entrants into the competing consolidator business could incur two types of barriers to entry: business implementation costs that emerge from the technical necessities of becoming a competing consolidator and regulatory compliance costs. The business implementation costs will include creation or modification of technical systems to receive, consolidate, and disseminate consolidated market data.[2035] Potential entrants will also need to satisfy the regulatory compliance requirements of Rule 614 to become competing consolidators, and many competing consolidators may need to eventually satisfy the regulatory requirements of

---

[2035] Competing consolidators will need to have systems and connections in place to receive data content from all SROs and then to disseminate the consolidated market data to a variety of market participants who will purchase their products. See supra Section V.C.1(b)(vi) and infra Section V.C.2(d). One commenter agreed that infrastructure costs would serve as a barrier to entry for potential competing consolidators. See NYSE Letter II at 15 ("[t]he significant costs required to develop, test, and support these technologies—costs that even existing data processors would incur—would serve as a barrier to entry for the competing consolidator market."). As discussed in detail in this section and below in Section V.C.2(d), the Commission believes that the costs for potential competing consolidators to develop and implement their systems will vary based on the type of entity that becomes a competing consolidator, but for some types of entities, these costs could be significantly higher and pose a larger barrier to entry.

Regulation SCI.[2036] Both the business implementation and regulatory compliance costs will differ based on the entrant type.[2037] The Commission believes that the barriers to entry will vary based on whether the potential competing consolidator is: a market data aggregation firm or a broker-dealer that currently aggregates market data for internal uses, one of the existing exclusive SIPs (which are operated by SROs), an SRO that does not operate an exclusive SIP, or a new entrant without experience aggregating market data. The business implementation costs will also vary based on the elements of consolidated market data the competing consolidator chooses to offer in their products.

The Commission believes that the existing market data aggregation firms and some broker-dealers that currently aggregate market data for internal uses could face low barriers to entry to become competing consolidators. Because they currently collect, consolidate, and, in some cases, disseminate market data to their customers, much like competing consolidators would, the Commission believes that firms and broker-dealers that currently aggregate proprietary market data would not have to extensively modify their systems. However, the Commission believes that each of these firms and broker-dealers would incur costs to expand their bandwidth and purchase hardware to receive information that is not currently disseminated in the exchange proprietary market data feeds, such as the regulatory data and administrative

---

2036 See supra Sections III.C.7 and III.C.8 (discussing the requirements of Rule 614). See also supra Section III.F (discussing the requirements of Rule 614(d)(9) and Regulations SCI). New entrants will face both initial implementation and ongoing costs to comply with these regulatory requirements. See infra Sections V.C.2(d) and V.C.2(e)(ii) (discussing these costs).

2037 See supra Sections IV.D.3 and IV.G. See also infra Sections V.C.2(d) and V.C.2(e)(ii).

data.[2038] Further, current market data aggregators and broker-dealers that currently aggregate market data for internal uses would incur new compliance costs to satisfy the regulatory compliance requirements to become competing consolidators, including costs associated with Form CC, as well as costs to comply with Rule 614(d)(9) and likely eventually Regulation SCI.[2039] These regulatory costs would initially be lower, but they could become large and therefore may affect entry and the benefits of the decentralized consolidation model. [2040]

The Commission believes that barriers to entry for a potential competing consolidator that is affiliated with an exchange—which could be one of the exclusive SIPs—would depend on several factors. In addition, both business implementation and regulatory compliance costs would be relatively lower for the existing exclusive SIPs than for the other competing consolidators that are affiliated with exchanges.

2038 See infra Section V.C.2(d).

2039 See id. See also infra Sections V.C.3 (for costs associated with Form CC); V.C.2(e)(ii) (for costs associated with Rule 614(d)(9) and Regulation SCI).

2040 Although potential competing consolidators will initially be subject to the lower costs of Rule 614(d)(9) rather than Regulation SCI, which will lower the initial barriers to entry, the Commission expects that many competing consolidators will eventually be SCI competing consolidators and that potential competing consolidators will take the higher costs of eventually becoming an SCI competing consolidator into account when deciding to enter the market. Rule 614(d)(9), which includes requirements similar to some of the key provisions of Regulation SCI, will apply to all competing consolidators (except competing consolidators affiliated with exchanges that do not operate under the limited exemptive relief) during the initial transition period and smaller competing consolidators that do not meet the market data revenue threshold for SCI competing consolidators thereafter. All competing consolidators that meet the consolidated market data revenue threshold for SCI competing consolidators, after the initial transition period, will be subject to Regulation SCI. See infra Section V.C.2(e)(ii) (discussing these costs). See also supra Section IV.G.3 (discussing number of competing consolidators subject to Regulation SCI).

The barriers to entry from business implementation costs to operate a competing consolidator would be relatively low for an exclusive SIP. Because the systems used by the exclusive SIPs already collect information in quotations and transactions from the SROs as well as aggregate and disseminate it, the exclusive SIPs would not have to make as extensive modifications to their systems as the other competing consolidators that are affiliated with exchanges.[2041] However, they would still incur costs to expand their bandwidth and connections to consume and disseminate consolidated market data as well as to transmit it with lower latency, and to program feed handlers to receive and normalize the different formats of the data feeds developed by the exchanges.[2042] On the other hand, the Commission believes that other competing consolidators that are affiliated with exchanges would likely have to build at least some new systems to process expanded core data, and thus, could incur relatively high initial implementation costs, though they may be able to keep their costs lower by leveraging some of their existing systems.[2043]

2041 Based on Commission staff experience, the Commission understands that existing exclusive SIPs' protocols for receiving direct data from exchanges are not standardized and introduce additional operational complexities. However, as the operators of exclusive SIPs, the exchanges, have figured out how to aggregate direct feeds for the purposes of their exchange matching engines, so they have the technology that would be deployable in the new decentralized consolidation model. If the exclusive SIPs determine to register as competing consolidators and to operate their competing consolidators using the existing infrastructure of the exclusive SIPs, then they may incur costs in order to reimburse each Plan's Participants for the costs they paid to build the exclusive SIP's systems. However, any determinations regarding payments to Participants or the disposition of the assets of the exclusive SIPs would be made by the Participants of the Equity Data Plans, subject to Rule 608. See supra note 979 and accompanying text.

2042 See supra Section V.C.1(b)(vi) and infra Section V.C.2(d).

2043 See supra Section IV.D.3 and infra Section V.C.2(d).

The barriers to entry from regulatory compliance costs would also be relatively lower for an exclusive SIP. Because the exclusive SIPs currently operate critical SCI systems,[2044] they will not bear any initial compliance costs associated with Rule 614(d)(9) and their ongoing compliance costs associated with Regulation SCI will not increase.[2045] SROs that do not operate exclusive SIPs are also already SCI entities. However, because these SROs do not have direct experience operating in the consolidated market data business, they may need to incur initial costs in order for their competing consolidator systems to be compliant with Rule 614(d)(9).[2046]

The other regulatory costs that the competing consolidators that are affiliated with exchanges would incur would vary based on whether they chose to operate under the provisions of the limited exemptive relief from the rule filing requirements of Section 19(b) of the Exchange Act and Rule 19b-4 thereunder, the denial of access provisions in Section 19(d) of the Exchange Act, the requirements in Section 6(b) of the Exchange Act, and from Regulation SCI in regard to their competing consolidators.[2047] However, the Commission believes that a

---

2044 See supra Section III.H.

2045 The Commission believes that the exclusive SIPs that become competing consolidators will likely surpass the 5% revenue threshold and will be required to comply with Regulation SCI at the end of the transition period, as described in the amendments. Their compliance costs associated with Regulation SCI may decrease, because the systems of an exclusive SIP that became a competing consolidator would no longer be considered critical SCI systems, which have stricter requirements and higher costs than other SCI systems. For example a critical SCI system needs to maintain backup systems that are designed to allow them to resume operations within two hours of a system outage (SCI entities only have the requirement to resume operations the day following a system outage). See infra V.C.2(e)(ii).

2046 See id.

2047 A competing consolidator affiliated with an exchange may be a facility of the exchange and subject to Section 19(b) of the Exchange Act and Rule 19b-4 thereunder. If a competing consolidator that is affiliated with an exchange chooses to act under the

competing consolidator that is affiliated with an exchange would choose to operate under the provisions of the limited exemptive relief because then they would not need to file rule changes (including new products and fee changes) related to their competing consolidator functions with the Commission under Section 19(b) of the Exchange Act.[2048] If these competing consolidators operate under the exemption, then they would still incur the other regulatory compliance costs associated with Rule 614.[2049] If these competing consolidators did not operate under the exemption, then they would need to comply with certain rules applicable to SROs, including the provisions of Regulation SCI and the requirements of Section 6(b), and to file all rule changes with the Commission under the Section 19(b) process, which would impose significant regulatory barriers in terms of making adjustments to their products and fees compared to other

---

limited exemptive relief, then the competing consolidator could do so pursuant to the conditions of the exemption and without having to operate under the denial of access provisions in Section 19(d) of the Exchange Act, the provisions of Regulation SCI related to an SRO (it would still be subject to the provisions of Regulation SCI related to competing consolidators), or without filing proposed rule changes with the Commission under Section 19(b) of the Exchange Act and Rule 19b-4 thereunder. Additionally, a competing consolidator that is affiliated with an exchange that chooses to operate under the limited exemptive relief would be exempt from the requirements of Section 6(b) of the Exchange Act (it would still be subject to the requirement in Rule 614(d)(3) to make consolidated market data products available to subscribers on terms that are not unreasonably discriminatory). See supra Section III.C.7(a)(iv).

2048 See id.

2049 A competing consolidator operating under the exemption would bear the regulatory compliance costs associated with Rule 614, including the costs associated with Form CC because the exemption requires the competing consolidator be registered as a competing consolidator under Rule 614 and be in compliance with the disclosure and other substantive regulatory requirements applicable to competing consolidators in Rule 603, Rule 614 and Form CC. Under the exemption, the exchange would also not be permitted to link the pricing for services of the affiliated competing consolidator to activities on, or other services performed by, the exchange. See id. See also infra Sections V.C.2(d) and V.C.3 for discussions of the regulatory compliance costs.

competing consolidators, potentially placing them at a competitive disadvantage.[2050] It would also create higher initial barriers to entry because the competing consolidator operations would need to be filed and approved by the Commission under Section 19(b) of the Exchange Act before they could begin operating.

The Commission anticipates that new entrants without prior experience in the market data aggregation business may become competing consolidators but that they would have the highest barriers to entry because they would incur both infrastructure and compliance costs. The new entrants would incur high infrastructure costs to build new systems to receive, consolidate, and disseminate consolidated market data; including costs to program feed handlers to be able to receive and normalize exchange data in different formats, and purchase bandwidth and connections to exchanges and co-location. These costs increase the fixed costs of participating as a competing consolidator in the market, further contributing to the barriers to entry. New entrants may also have the highest compliance costs among all potential entrants, because they would have to build compliance systems from scratch to satisfy both Rule 614(d)(9), and later potentially Regulation SCI, as well as the other requirements of Rule 614, including Form CC. Therefore, the Commission believes that there may be a limited number of firms that could enter the market data aggregation business for the first time.

The business implementation compliance costs will vary based on the elements of consolidated market data the competing consolidator chooses to offer in their products.

[2050] Rule filings under Section 19(b) would be subject to a notice and comment process and Commission consideration. Fee changes could be immediately effective upon filing under Section 19(b)(3), but the Commission would have the authority to abrogate such fee changes.

Specifically, potential entrants that seek to specialize in offering data products to clients who do not wish to receive the full consolidated market data could save on ongoing costs and potentially also on initial infrastructure costs.[2051] The initial cost savings would vary across the entrant types listed above depending on the extent to which the entrant has already built the infrastructure necessary to aggregate and distribute data similar to consolidated market data. For example, current data aggregators choosing to specialize are likely to see a small reduction in barriers to entry from this change while firms without prior experience are likely to see a significant reduction in barriers to entry.

One commenter stated that the Commission did not consider the risks of the potential liability that a competing consolidator may incur for any performance failures, which are a significant barrier to entry.[2052] The Commission believes that these potential liability concerns are not a significant barrier to entry for competing consolidators. Competing consolidators could attempt to limit their potential liability from systems issues through contractual agreements with their subscribers, similar to provisions that data providers currently include in their subscriber agreements.[2053]

2051 See supra Section III.C.1(b) and infra Section V.C.2(d)(i).

2052 See NYSE Letter II at 15.

2053 See, e.g., CTA Plan Professional Subscriber Agreement, available at https://www.ctaplan.com/publicdocs/ctaplan/notifications/trader-update/Professional%20Subscriber%20Agreement.pdf; UTP Plan Subscriber Agreement, available at http://www.utpplan.com/DOC/subagreement.pdf; Nasdaq Global Subscriber Agreement, available at: http://www.nasdaqtrader.com/content/AdministrationSupport/AgreementsData/subagreemstandalone.pdf; ICE Data Services and Software Services Agreement, available at: https://www.theice.com/publicdocs/agreements/ICE_Data_Services_Agreement.pdf.

b. Effective National Market System Plan(s) Fees for Data Content Underlying Consolidated Market Data

Another factor that would affect the number of competing consolidators relates to the fees that the effective national market system plan(s) would set for the consolidated market data content.[2054] If these fees are set too high or have the effect of limiting product differentiation,[2055] they could limit the opportunities for competing consolidators to build profitable businesses.

The Commission recognizes uncertainty in these fees. The fees developed by the effective national market system plan(s) for the data content underlying consolidated market data offerings would be proposed by the Operating Committee(s) of the national market system plan(s) and filed with the Commission.[2056] Because such fees depend on future action by the effective national market system plan(s), the Commission cannot be certain of the level of those fees or whether such fees would provide discounts for those end users who wish to receive subsets of consolidated market data (e.g., different prices for different levels of data content or different core data component) or based on usage categories (e.g., professional, non-professional, non-display).[2057] As discussed further below, the fees developed by the Operating Committee of the effective national market system plan(s) must be fair and reasonable and not unreasonably discriminatory.

2054 See infra Section V.C.2(b) for a discussion on the economic analysis of data content, consolidation and dissemination, and connectivity fees.

2055 See infra Section V.C.2(a)(i)e for a discussion on the potential dimensions of product differentiation by competing consolidators.

2056 See Proposing Release, 85 FR at 16837.

2057 See infra Section V.C.2(b)(ii) for further discussion of the impact of providing discounts based on scope of data content.

Some commenters said that the uncertainty over fees for data content underlying consolidated market data offerings will make it difficult for potential competing consolidators to estimate the economic value of this new business opportunity, and therefore, enter the market.[2058] While there is uncertainty surrounding the currently unknown levels of data content fees, potential competing consolidators can judge the value of the business opportunity. Potential competing consolidators will see, and be able to comment on, the newly proposed data content fees before they will have to decide whether to register as competing consolidators.[2059] During the transition period, the new data content fees proposed by the effective national market system plan(s) will be available for potential competing consolidators to review and comment on before the registration date for the initial competing consolidator wave expires. This will give competing consolidators adequate time to evaluate this information and the potential business opportunity.

Additionally, the Commission believes that there will likely be different levels of fees for data content underlying consolidated market data offerings either based on usage category (e.g., professional, non-professional, non-display) or based on the scope of data content market participants use or a combination of both.[2060] In either case, the Commission believes that the differential pricing of consolidated market data will expand differentiation opportunities for the potential competing consolidator, as discussed below.[2061]

2058 See, e.g., NYSE Letter II at 14.

2059 See supra Section III.H for a discussion of the steps during the transition period.

2060 See infra Section V.C.2(b) for a discussion in the impact on data fees.

2061 See infra Section V.C.2(a)(i)e for a discussion on competing consolidators' differentiation.